REV12 - PROJECT ENGINEERING PIPELINE

Completed architecture, remaining engineering tasks, and release gates for electrical cabinet construction.

ARCHITECTURE DEFINED — **DETAILED DESIGN + VALIDATION + RELEASE STILL REQUIRED**

Architecture	Device allocation	Detailed wiring	Safety validation	PLC / HMI build	Cabinet release	Build + FAT
COMPLETE	COMPLETE	IN PROGRESS	REQUIRED	REQUIRED	HOLD	AFTER RELEASE

COMPLETED / DEFINED

System architecture

PLC, SCALANCE XB116, IFM AL1403 masters and field-device topology documented.

IO-Link port maps

AL102 / AL104 physical ports shown with device allocation and connection path.

Encoder architecture

IFM RO3101 assigned to Siemens TM Count 2x24V, 6ES7550-1AA01-0AB0.

Pilz safety architecture

PNOZmulti 2 modules, PSEN door switches and PROFINET diagnostic boundary documented.

Stack light

Leuze TL100 command path defined through AL102 X8 and IFM AL2330 X1 outputs.

Machine work lights

Four 24 VDC / 40 W lights; Siemens 3RT2016-1KB41 contactor selected.

REMAINING ENGINEERING WORK

1 M104 hood fan

Nameplate confirmed: Nidec, 310 VDC, 58 W, 1350 rpm, 8-pole. Obtain control wiring diagram.

2 Light-circuit release

Contactor selection CLOSED. Verify LED inrush, PSU, branch fuses, conductors and terminals.

3 Sensors + IO-Link

Confirm pressure-sensor model and pinout. Approve LMT100 SIO / IO-Link parameter set.

4 Pendant / JOG

Issue final pinout, contact allocation and verified PLC addresses from product documentation.

5 Pilz safety

Complete risk assessment and PNOZmulti project. Allocate 11 doors, E-stops, STO, EDM and restart.

6 Networks + I/O

Assign SCALANCE ports, IP plan and cable IDs. Close AL102 / AL104 spare or planned allocations.

CABINET CONSTRUCTION RELEASE GATE

Release only after approved schematics, final BOM, cable / terminal schedule, load and protection calculations, safety validation, PLC / HMI I/O review, and issued-for-construction sign-off.